Food-Composition Foundation Evidence under Explicit Dependency Masking: Revised Analyses

Version: 2026-10-28 r2

Primary model: V5+CCE (dependency-mask v1)

Scope: mask audit, food/database UMAP, Taiwan zero/few-shot, sparse completion, INFOODS group sensitivity, dependency exposure ablation

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Executive summary

The dependency mask affects both training and inference. Every training profile-target task hides the target plus its directed ancestors and descendants; validation, complete test, Taiwan transfer, and all analyses use the same policy. V4, V5, and V5+CCE were trained from scratch under this mask.

This revision keeps only food-category and database-source UMAP views; compares frozen V5+CCE with Taiwan few-shot RF/XGBoost without a scratch Transformer; restricts sparse completion to V5+CCE, RF, XGBoost, and KNN; and adds an FAO/INFOODS component-group sensitivity matrix plus a dependency exposure ablation.

Key results:

- Frozen V5+CCE scores 0.2000 on the fixed Taiwan test. RF scores 0.4551/0.3266/0.2430 at 8/32/128 shots; XGBoost scores 0.5287/0.3262/0.2369. The frozen model is better at every budget.
- For +10% input perturbations, within-group prediction shift is 0.00550 versus 0.00179 between groups; deletion gives 0.08410 versus 0.02230.
- The inference-only dependency exposure ablation does not show stronger use of strict dependency relatives than ordinary controls. Exposing all normally hidden relatives changes SLMAE from 0.1541 to 0.1628 and therefore worsens accuracy.

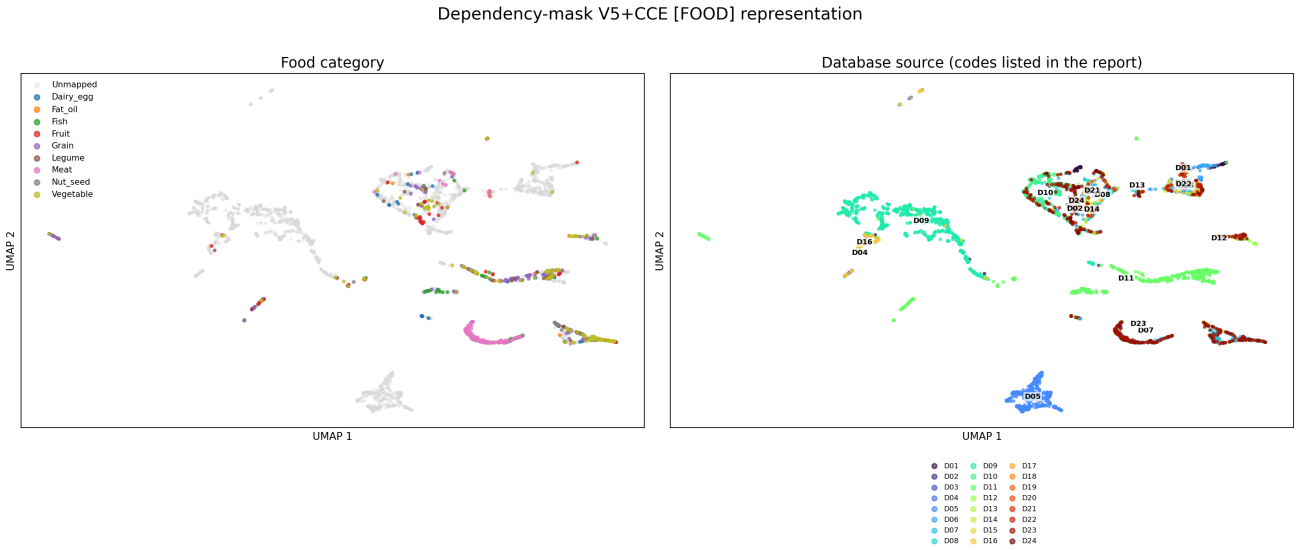
1. Shared setup and mask

The graph contains only totals, components, subtotals, and structural members explicitly supported by FAO/INFOODS tagname definitions. It has 37 directed edges over 44 axes; 144 of 187 supervised targets are singletons. A target mask is `target + all directed ancestors + all directed descendants`; siblings remain visible and the legacy `mask_family` column is unused.

Method	SLMAE	SLRMSE	Raw MAE
V5+CCE	0.3016	0.4910	0.3137
XGBoost	0.3270	0.4825	0.2624
RF	0.3322	0.4841	0.2838
KNN	0.4047	0.5937	0.4729

2. UMAP: food category and database

UMAP uses the first-pass V5+CCE `[FOOD]` representation for 4,000 validation profiles. Category and database labels are used only for colouring. Observation-count and single-nutrient gradient panels have been removed.



Food-category and database UMAP

Representation	Macro purity@10	Macro-F1	Balanced acc.
Values+mask	0.7563	0.8614	0.8598
V4	0.6424	0.7296	0.7505
V5	0.6767	0.7592	0.7814
V5+CCE	0.6973	0.7895	0.8096

Database codes are listed below.

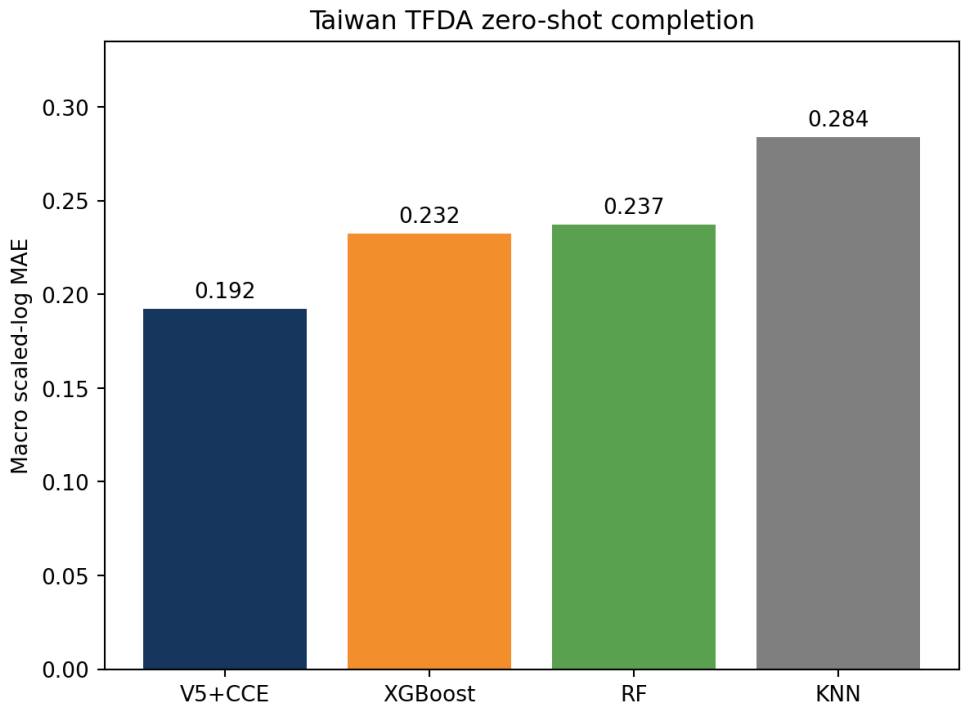
Code	Source key	Database	UMAP n
D01	afcd	Australian Food Composition Database	107
D02	anfood_2_0	FAO/INFOODS AnFooD 2.0	22
D03	bangladesh_fct_2013	Bangladesh Food Composition Table 2013	15
D04	biofoodcomp_4_0	FAO/INFOODS BioFoodComp 4.0	51
D05	bls_4_0	German Nutrient Database BLS 4.0	405
D06	ciqual	ANSES-Ciqual	194
D07	cnf	Canadian Nutrient File	326
D08	cofid	UK CoFID	141
D09	efsa_eu_fcdb_2013	EFSA EU Food Composition Database	830
D10	fndds	USDA FNDDS	273
D11	foodb	FooDB	622
D12	frida	Frida FoodData, Denmark	79
D13	lesotho_fct_2006	Lesotho Food Composition Table 2006	12
D14	mext_japan_2023	Japan MEXT Food Composition Tables	133
D15	norway	Norwegian Food Composition Database	87
D16	phyfoodcomp_1_0	FAO/INFOODS/IZiNCG PhyFoodComp 1.0	112
D17	smiling_cambodia_2013	SMILING Cambodia FCT 2013	6

Code	Source key	Database	UMAP n
D18	smiling_indonesia_2013	SMILING Indonesia FCT 2013	6
D19	smiling_laos_2013	SMILING Laos FCT 2013	6
D20	smiling_thailand_2013	SMILING Thailand FCT 2013	7
D21	smiling_vietnam_2013	SMILING Vietnam FCT 2013	11
D22	swiss_fcdb_7_1	Swiss Food Composition Database 7.1	73
D23	usda_sr_legacy	USDA Standard Reference Legacy	431
D24	wafct_2019	FAO/INFOODS West Africa FCT 2019	51

Missingness alone reaches about 0.971 source balanced accuracy, so database separation is materially confounded by coverage patterns.

3. Taiwan zero-shot

Taiwan is excluded from training, scale estimation, tuning, and checkpoint selection. The strict panel contains 2,212 foods, 128,470 food-axis jobs, and 80 axes.



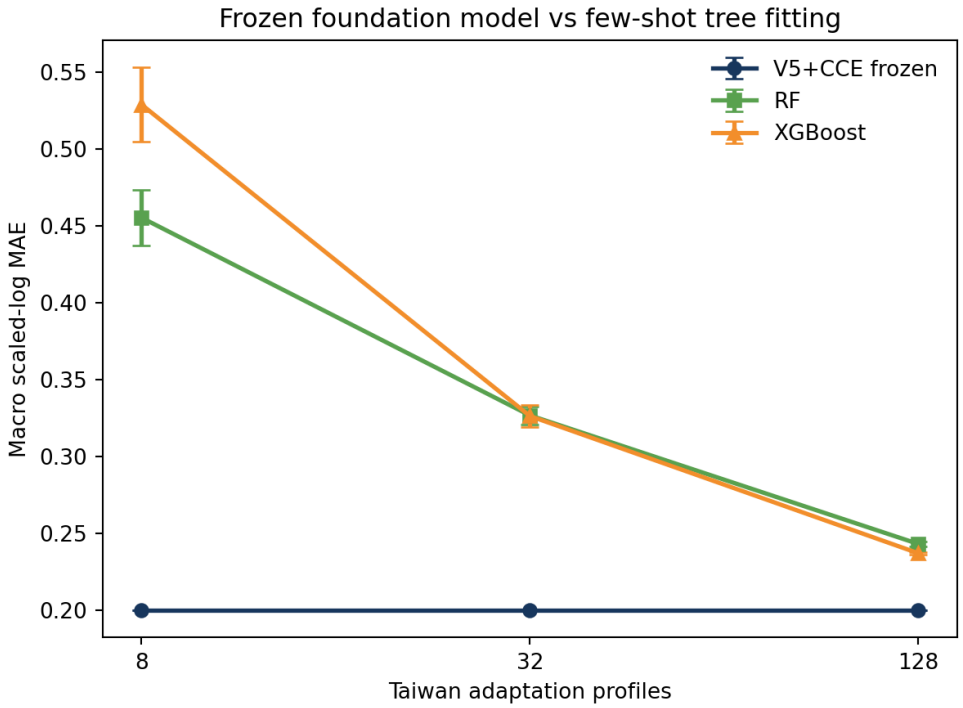
Taiwan zero-shot

Method	SLMAE	SLRMSE	Raw MAE
V5+CCE	0.1922	0.3703	0.3253
XGBoost	0.2324	0.3731	0.3185
RF	0.2370	0.3729	0.3332
KNN	0.2838	0.4519	0.4968

V5+CCE has the best primary scaled-log MAE, while raw MAE can still favour XGBoost.

4. Taiwan few-shot: frozen model versus tree fitting

A fixed 400 name-novel food test is held out from 1,389 adaptation foods. The 8-shot subset first uses missingness-only set coverage so every test axis has at least one label, then fills randomly; subsets are nested through 32 and 128. V5+CCE remains frozen. RF and XGBoost use their formal configurations and are fit per axis only on the selected Taiwan profiles. There is no scratch Transformer and no Transformer fine-tuning.



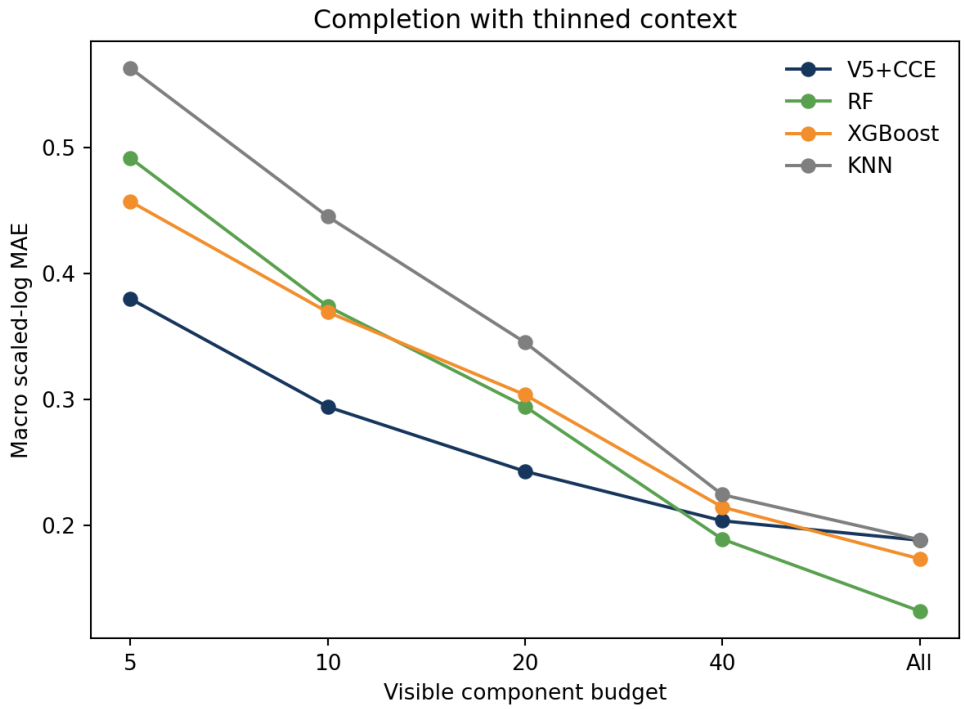
Taiwan few-shot

Method	Profiles	SLMAE mean	SD
RF	8	0.4551	0.0179
V5+CCE frozen	8	0.2000	0.0000
XGBoost	8	0.5287	0.0243
RF	32	0.3266	0.0058
V5+CCE frozen	32	0.2000	0.0000
XGBoost	32	0.3262	0.0074
RF	128	0.2430	0.0017
V5+CCE frozen	128	0.2000	0.0000
XGBoost	128	0.2369	0.0010

Even at 128 Taiwan foods, XGBoost reaches 0.2369 and RF 0.2430, compared with 0.2000 for frozen V5+CCE. This directly demonstrates external sample efficiency from pretraining.

5. Completion with few visible observations

The panel fixes 12 high-support targets and 200 validation jobs per target. Dependency masking is applied first, followed by random retention of 5, 10, 20, 40, or all context values. Only the four requested methods are retained.



Sparse completion

Method	Visible	SLMAE mean	SD
KNN	5	0.5633	0.0069
RF	5	0.4918	0.0077
V5_CCE	5	0.3799	0.0078
XGBoost	5	0.4572	0.0043
KNN	10	0.4453	0.0029
RF	10	0.3738	0.0099
V5_CCE	10	0.2940	0.0042
XGBoost	10	0.3693	0.0086
KNN	20	0.3452	0.0025
RF	20	0.2943	0.0021
V5_CCE	20	0.2427	0.0061
XGBoost	20	0.3035	0.0020
KNN	40	0.2241	0.0022
RF	40	0.1888	0.0027
V5_CCE	40	0.2033	0.0018
XGBoost	40	0.2142	0.0005
KNN	all	0.1883	nan
RF	all	0.1316	nan
V5_CCE	all	0.1879	nan
XGBoost	all	0.1731	nan

V5+CCE leads at 5, 10, and 20 visible values. RF leads at 40 and all, with XGBoost second. The foundation-model advantage is concentrated in genuinely sparse context.

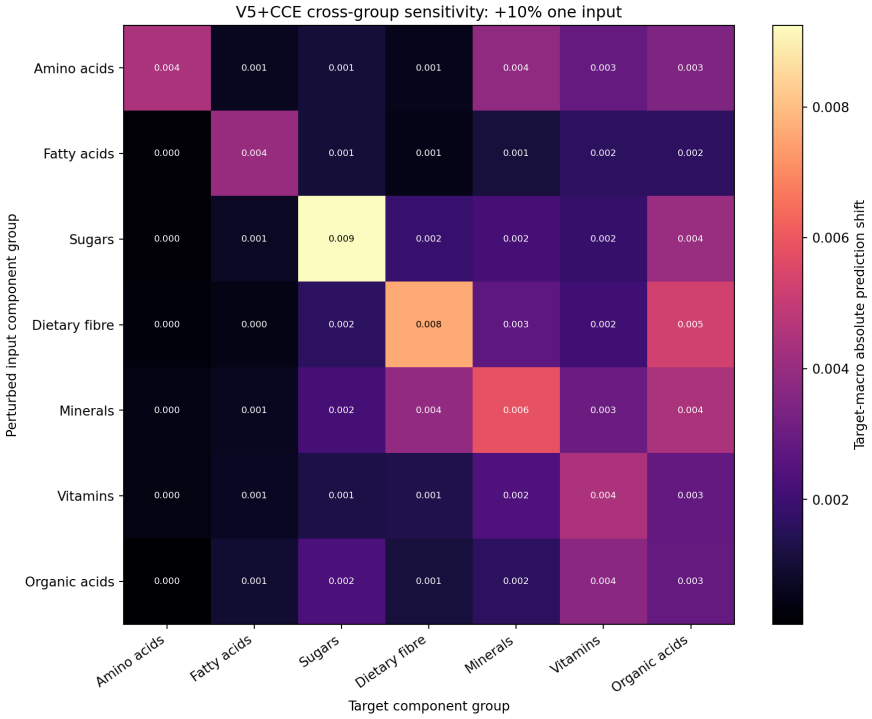
6. FAO/INFOODS component-group sensitivity

6.1 Classification authority

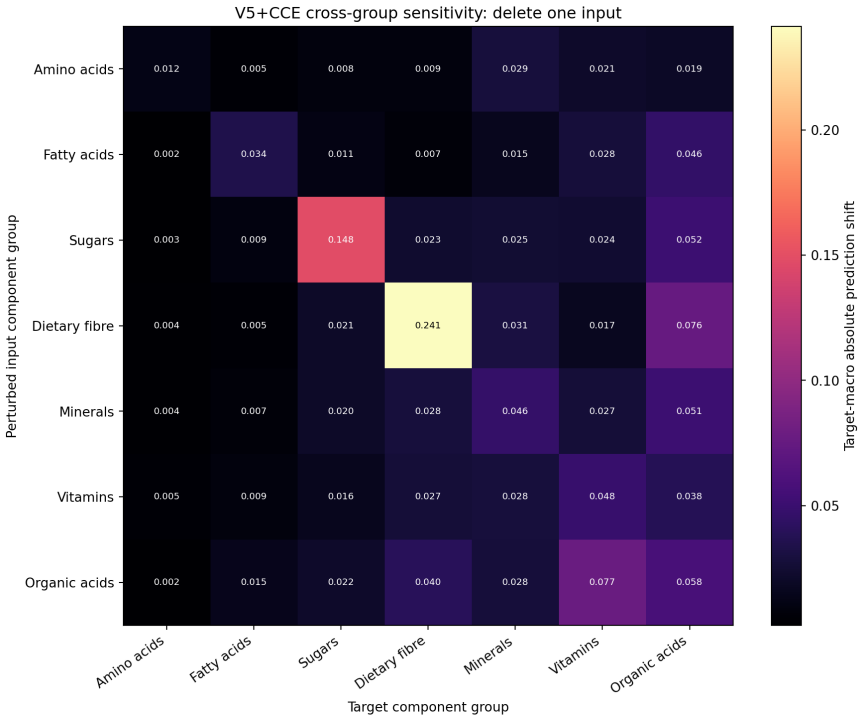
One source is used: the FAO/INFOODS and EuroFIR component nomenclature workshop report, Annex 1, with names checked against INFOODS tagnames. The Annex hierarchy is labelled a draft proposition, so only seven clear groups are registered: amino acids, fatty acids, sugars, dietary fibre, minerals, vitamins, and organic acids. Unlisted axes remain unclassified. The mapping covers 135/252 axes and 132 supervised targets. It is used only for sensitivity aggregation and never for masking or model input.

6.2 Cross-group matrix

After dependency masking, one positive visible input is selected from every available input group for each target job. The audit contains 58,197 input-target pairs.



+10% group sensitivity



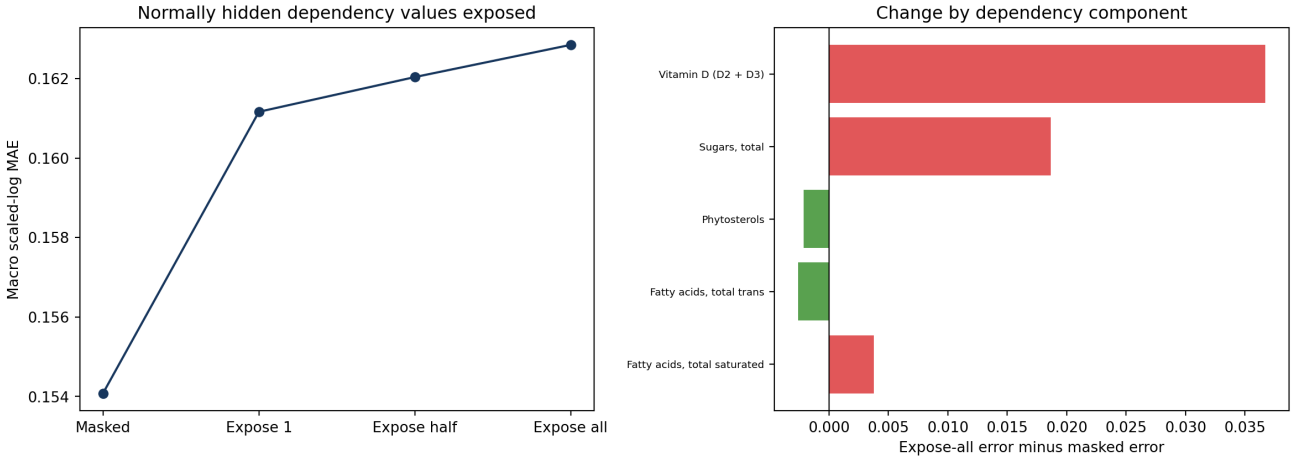
Single-input deletion group sensitivity

Condition	Relation	Prediction shift	Error delta	Pairs
deletion	between_group	0.0223	0.0013	45269
deletion	within_group	0.0841	0.0349	12928
minus_10pct	between_group	0.0020	-0.0002	45269
minus_10pct	within_group	0.0061	-0.0002	12928
plus_10pct	between_group	0.0018	0.0002	45269
plus_10pct	within_group	0.0055	0.0010	12928

The diagonal is about 3.1 times the off-diagonal mean for +10% and 3.8 times for deletion. Sugars and dietary fibre have the strongest diagonal responses, although fibre cells have low support. These are predictive associations, not causal nutritional interventions.

7. Dependency-family exposure and perturbation ablation

This experiment uses only target-specific ancestors/descendants that the current policy normally hides; siblings are already visible. It covers 6,889 jobs, 32 targets, and 5 dependency components.

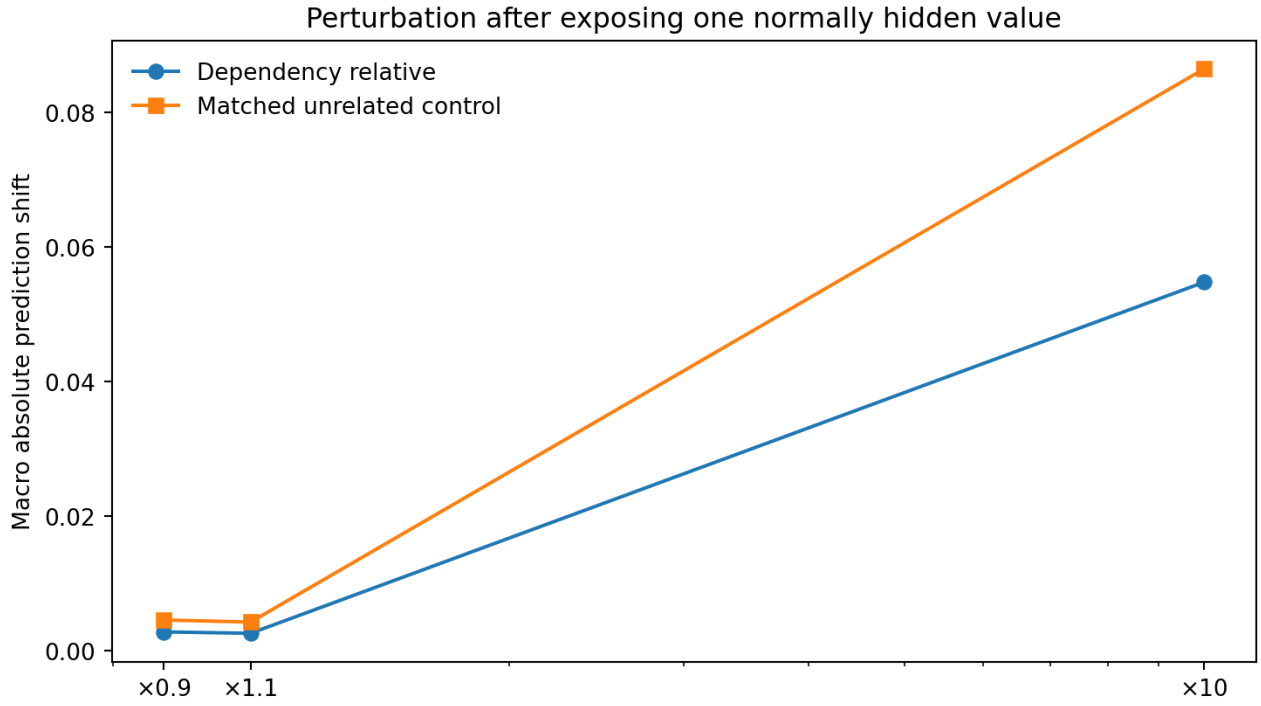


Dependency exposure ablation

Condition	SLMAE	Shift vs masked	Axes	Jobs
standard_mask	0.1541	0.0000	32	6889
expose_one	0.1612	0.0237	32	6889
expose_half	0.1620	0.0256	32	6889
expose_all	0.1628	0.0290	32	6889

Exposing more dependency values does not improve accuracy. It is an out-of-distribution inference condition because training never allowed a target to see these relatives.

One exposed dependency value and a matched unrelated visible control are then perturbed separately.



Dependency versus matched-control perturbation

Factor	Dependency shift	Control shift	Dependency Δerror	Control Δerror
0.9000	0.0028	0.0046	-0.0007	0.0003
1.1000	0.0026	0.0042	0.0007	0.0004

Factor	Dependency shift	Control shift	Dependency Δerror	Control Δerror
10.0000	0.0548	0.0865	0.0177	0.0400

Revealing one dependency value shifts prediction by 0.02371, versus 0.02991 for the control. The profile-bootstrap 95% interval for dependency minus control is [-0.01124, 0.00091]. Dependency perturbation shifts are also lower overall at all three factors.

This ablation therefore does not establish that the trained model strongly uses these strict parent-child relations when they are exposed only at inference. It supports retaining the mask because inference-only unmasking worsens error. The scientific relation itself comes from INFOODS definitions. Quantifying leakage from omitting the mask would require a separately trained matched no-dependency-mask model.

8. Conclusions and limitations

1. The new mask is part of training and every evaluation path.
2. Frozen V5+CCE beats RF/XGBoost fit on 8–128 labelled Taiwan foods, supporting external sample efficiency.
3. V5+CCE leads sparse completion at 5–20 visible values; trees lead with richer context.
4. Broad INFOODS groups show stronger within-group than between-group response, but this is associative.
5. Inference-only strict-dependency exposure is a negative result and cannot prove learned dependency use.
6. UMAP database separation is strongly confounded by missingness; Taiwan covers 80/187 targets; neural evidence uses one pretrained seed.

9. Reproducibility and attachments

The package contains aggregate tables, figures, hashes, and scripts. It excludes checkpoints, raw data, and row-level predictions.